

VIANA DO CASTELO-CHAFE, Portugal

WMO: 085510

Lat: 41.6489N

Lon: 8.8047W

Elev: 49

StdP: 100.74

Time Zone: 0.00 (EUW)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.0	2.2	-3.5	2.8	6.1	-1.6	3.3	5.7	8.2	11.2	6.7	12.9	2.2	100	0.403

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	8.8	29.2	18.8	27.0	18.0	25.2	17.8	20.5	25.1	19.7	23.6	19.2	22.7	2.5	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.3	14.2	21.3	18.7	13.6	20.6	18.1	13.1	20.1	59.1	25.0	56.5	23.7	54.7	22.8	24.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.5	5.4	4.7	-0.9	33.4	1.3	1.8	-1.8	34.7	-2.6	35.7	-3.3	36.8	-4.2	38.1		
(5)			-1.7	21.9	1.1	1.3	-2.5	22.9	-3.2	23.6	-3.8	24.3	-4.6	25.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	14.5	9.3	10.0	11.7	13.5	15.6	17.6	19.0	19.2	18.5	16.1	12.3	10.3		
	DBStd	4.38	2.59	2.36	2.32	2.76	2.76	2.50	2.24	2.34	2.52	2.75	2.80	2.80		
	HDD10.0	125	44	27	9	2	0	0	0	0	0	1	10	32		
	HDD18.3	1566	279	233	206	147	94	42	14	13	27	78	181	251		
	CDD10.0	1751	23	27	61	107	174	229	280	286	254	191	79	39		
	CDD18.3	152	0	0	0	2	10	21	36	41	32	10	0	0		
	CDH23.3	895	0	0	2	17	88	135	216	233	166	38	0	0		
	CDH26.7	217	0	0	0	2	21	34	64	58	35	3	0	0		
(14) Wind	WSAvg	2.1	2.0	2.3	2.4	1.9	1.9	1.9	2.0	2.0	1.9	1.9	1.9	2.6		
(15) Precipitation	PrecAvg	1422	190	170	110	112	102	53	24	27	91	156	176	211		
	PrecMax	1906	432	482	346	349	255	189	81	97	366	533	492	583		
	PrecMin	858	18	5	0	20	4	0	0	0	1	7	0	28		
	PrecStd	268	106	109	78	73	61	36	23	27	79	103	112	124		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.0	19.2	22.7	25.5	29.3	30.1	31.8	31.1	30.2	26.8	21.2	18.5		
		MCWB	12.2	12.9	12.9	15.3	17.1	18.9	19.2	20.3	19.0	18.0	16.2	12.5		
	2%	DB	14.7	16.5	19.9	22.7	25.8	27.3	28.3	28.1	27.1	24.1	18.9	16.5		
		MCWB	13.2	11.8	12.5	15.0	16.3	18.3	18.9	18.9	17.8	16.8	15.5	13.5		
	5%	DB	14.1	15.0	17.6	20.6	23.0	24.4	25.5	25.6	25.1	21.9	17.5	15.5		
		MCWB	12.9	11.8	11.8	14.7	16.1	17.8	18.7	18.3	17.5	16.6	15.2	13.6		
	10%	DB	13.4	14.0	15.9	18.6	20.8	22.4	23.5	23.9	23.2	20.2	16.4	14.6		
		MCWB	12.3	11.7	11.6	14.0	15.7	17.2	18.2	18.2	17.6	16.8	14.6	13.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.6	14.5	14.9	16.9	19.0	21.0	21.2	22.1	20.7	19.8	18.0	16.1		
		MCDB	14.9	15.6	17.8	22.2	25.0	27.5	27.1	27.0	25.9	22.3	19.0	16.3		
	2%	WB	13.9	13.6	14.1	16.1	17.7	19.5	20.1	20.6	19.7	19.1	16.7	15.0		
		MCDB	14.1	14.8	16.1	20.8	22.9	24.1	24.9	24.1	23.8	20.7	17.7	15.4		
	5%	WB	13.2	12.9	13.4	15.3	16.9	18.6	19.4	19.7	19.0	18.4	16.0	14.2		
		MCDB	13.6	14.2	15.4	19.3	21.3	22.6	23.5	23.1	22.3	20.0	17.1	14.9		
	10%	WB	12.6	12.1	12.7	14.6	16.2	17.8	18.8	19.0	18.3	17.7	15.1	13.4		
		MCDB	13.2	13.7	14.9	17.8	19.7	21.4	22.5	22.2	21.5	19.4	16.1	14.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.7	7.7	8.3	8.3	8.4	8.1	8.3	8.8	9.0	8.1	7.0	7.1		
		MCDBR	6.0	9.4	11.8	11.6	12.2	11.2	11.6	11.7	11.7	10.5	7.7	7.0		
	5% WB	MCWBR	4.7	5.8	6.6	5.8	5.7	5.0	4.9	5.1	4.9	5.2	5.0	5.1		
		MCWBR	3.8	7.2	8.0	9.4	9.6	9.1	9.0	9.0	9.0	6.9	5.7	5.2		
(40) Clear-Sky Solar Irradiance	taub	taub	0.338	0.346	0.381	0.379	0.386	0.389	0.378	0.390	0.378	0.372	0.344	0.333		
		taud	2.480	2.446	2.355	2.372	2.379	2.389	2.411	2.378	2.427	2.455	2.486	2.487		
	Ebn at Noon	Ebn at Noon	808	859	864	891	890	885	891	870	857	817	792	780		
		Edh at Noon	76	91	111	117	119	118	114	115	102	89	75	70		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	1.76	2.79	3.91	5.21	6.33	6.89	6.98	6.31	4.92	3.15	2.04	1.59		
	RadStd	RadStd	0.30	0.39	0.58	0.56	0.54	0.46	0.39	0.37	0.34	0.38	0.34	0.24		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+36

Nomenclature: See separate page